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Optimization and analysis of porosity and roughness in selective laser melting 316L parts

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ABSTRACT

Purpose: The investigations have been carried out on 316L stainless steel parts fabricated by Selective Laser Melting (SLM) technique. The study aimed to determine the effect of SLM parameters on porosity, hardness, and structure of 316L stainless steel.

Design/methodology/approach: The analyses were conducted on 316L stainless steel parts by using AM125 SLM machine by Renishaw. The effects of the different manufacturing process parameters as power output, laser distance between the point's melted metal powder during additive manufacturing as well as the orientation of the model relative to the laser beam and substrate on porosity, hardness, microstructure and roughness were analysed and optimised.

Findings: The surface quality parts using 316L steel with the assumed parameters of the experiment depends on the process parameters used during the SLM technique as well as the orientation of formed walls of the model relative to the substrate and thus the laser beam. The lowest roughness of 316L SLM parts oriented perpendicularly to the substrate was found when 100 W and 20 μm the distance point was utilised. The lowest roughness for part oriented at 60° relatives to the substrate was observed when 125 W and the point distance 50 μm was employed.

Practical implications: Stainless steel is one of the most popular materials used for selective laser sintering (SLM) processing to produce nearly fully dense components from 3D CAD models. Reduction of porosity is one of the critical research issues within the additive manufacturing technique SLM, since one of the major cost factors is the post-processing.

Originality/value: This manuscript can serve as an aid in understanding the importance of technological parameters on quality and porosity of manufactured AM parts made by SLM technique.

Keywords: Manufacturing process parameters, Additive manufacturing, SLM, Porosity, Roughness, Microstructure, 316L

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PROPERTIES

1. Introduction

The era of 3D printing opened many new possible ways to produce elements with more complex shapes in cheaper and faster way. Now it is possible to print parts of the human body such as a kidney. Any different technology doesn't allow to do it, only 3D printing. It is possible to make almost whole car or plan using additive technologies. Nowadays everything is possible, but many people don't know how enormous potential is in 3D printing. In manufacturing, usually is used metal powders or polymers to produce parts but in a few years, it will be possible to use ceramics to create components in colossal scale. In medicine will occur enormous progress. It will be able to use hydroxyapatite to produce bone parts. In each of 3D printing technology, is possible to change parameters of the process. That allows manipulating the properties of the printed piece. Each application may require other features. For one item more critical will be higher hardness and higher porosity for another. By changing production parameters, it is possible to obtain the most attractive properties. Choosing the best combination of production parameters will save time and money during further stages of production [1-7]. Figure 1 shows a brief timeline of 3D printing. Nowadays, in possession of engineering exist many different types of 3D printers that use various techniques and different materials.

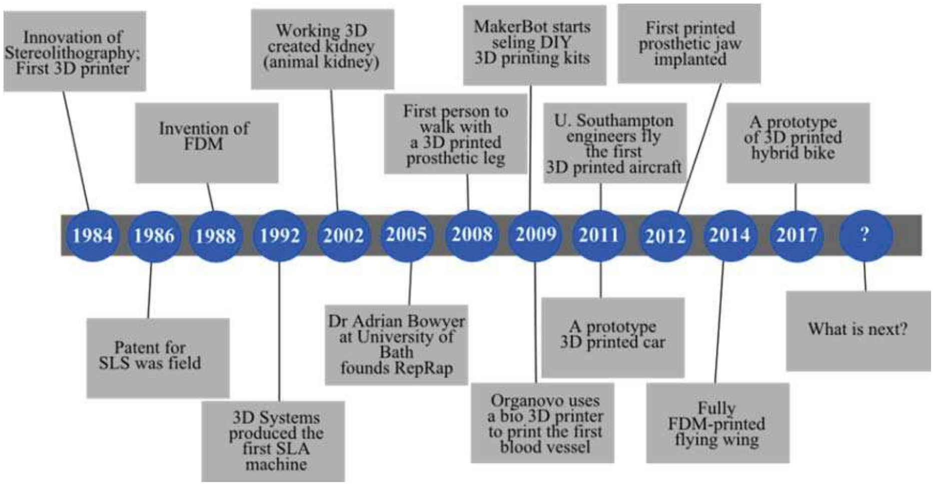


Fig. 1. 3D printing timeline [1-3]

Nowadays, the key determinants of competitiveness in the market are the price of the product offered, its quality and the time it will occur on the market. Aspects such as fashion trends, cultural habits, and social conditions are also important. Customer feedback on the product is a significant consideration when deciding to launch a new product on the market. Diversification of customer needs forces manufacturers to continuous improvement and be flexible in rebuilding their stockpiles with the tools they need. Companies must be prepared to shorten production cycles with minimal investment outlay. To meet these tasks, companies are forced to use the right tools [8-14].

Multiple prototypes of physical models and their various improvements are now possible through rapid prototyping (RP) and rapid tooling (RT) methods. These technologies are techniques of rapidly producing physical models of products and their components. They also allow the production of technical, functional and visual prototypes without the use of conventional machining techniques [15-20].

Before the production of an SLM parts, many fundamental geometrical features such as the point sequence, the border geometry, the hatch geometry, and the support geometry, must be defined. The proper optimisation of all these features is carried out by taking into consideration several properties such as the part's surface finish and microstructure, its mechanical properties, and the production time and cost. However, the essential target when introducing new material to the process is the production of sufficiently dense parts.

This work focus on investigating the effect of changes in parameters (laser power and point distance) and orientation of manufactured parts achieved by selective laser melting on selected properties such as porosity, hardness, microstructure and roughness of 316L steel parts.

2. Materials and methods

Renishaw's 316L stainless steel powder was used for the samples preparation. This material is one of the most widely used austenitic stainless steels, among others in the chemical, petrochemical, pharmaceutical, home appliance and many other industries where excellent corrosion resistance is required. This steel does not show any change in operation, and the secretion effects can only appear after several hours of tempering. The particle size of the powder ranged from 15 to 45 µm. The chemical composition of the 316L powder used in the experiment is given in Table 1.

The selected technological properties of powder 316L like flowability, bulk density and tapped density were measured by Hall Flowmeter funnel according to ISO 3923-2.

Table 1.
Chemical composition of the material used in processing

Element	wt.%	at.%
Si	1.6	3.1
Mo	2.6	1.5
Cr	19.8	21.0
Mn	3.1	3.1
Fe	61.7	60.8
Ni	11.0	10.3

Renishaw's AM 125 machine was used to produce the samples by selective laser melting. The source of radiation is YFL continuous wave Ytterbium fibre laser with a wavelength λ=1070 nm. To determine the influence of manufacturing processing parameters on the structure, hardness, porosity and roughness of parts, a matrix with different processing parameters was created (Table 2).

Energy density is typically used to define the average applied energy per volume of material during a powder bed fusion process. Essential processing parameters are included in an energy density equation. For example, energy density can be expressed by equation (1) for the SLM process [11].

$$E = \frac{P}{v \cdot h \cdot s} \left[\frac{J}{mm^3}\right] \quad (1)$$

where P is laser power, v is scanning speed, h is point distance, and s is layer thickness. Higher laser power and lower point distance, scan speed and layer thickness increase the amount of energy, which induces heat into the powder to melt it (Table 3). In this study, the influence of the volume energy density on the porosity, roughness, structure and hardness is studied by the variation of the two different parameters: laser power and scan speed. Others manufacturing parameters were constant, i.e. layer thickness – 50 µm, exposure time – 100 µs, spot diameter – 35 µm, laser scanning speed – 480 mm/s.

The manufactured samples consisted of two walls, one of which was oriented perpendicular to the substrate during manufacturing and the other at an angle of 60° (Figure 2).

Table 2.
Matrix of variable parameters

		Point distance, µm / Print angle, °				
		10	20	30	40	50
Output power, W	100	90/60	90/60	90/60	90/60	90/60
	125	90/60	90/60	90/60	90/60	90/60
	150	90/60	90/60	-*	90/60	90/60
	175	90/60	90/60	90/60	90/60	90/60

*Sample was destroyed during the preliminary tests as a result of equipment failure for mounting samples which made it impossible to use them for research.

Table 3.
Matrix of calculated energy density, J/mm^3

		Point distance, µm				
		10	20	30	40	50
Output power, W	100	417	208	139	104	83
	125	521	260	174	130	104
	150	625	313	-	156	125
	175	729	365	243	182	146

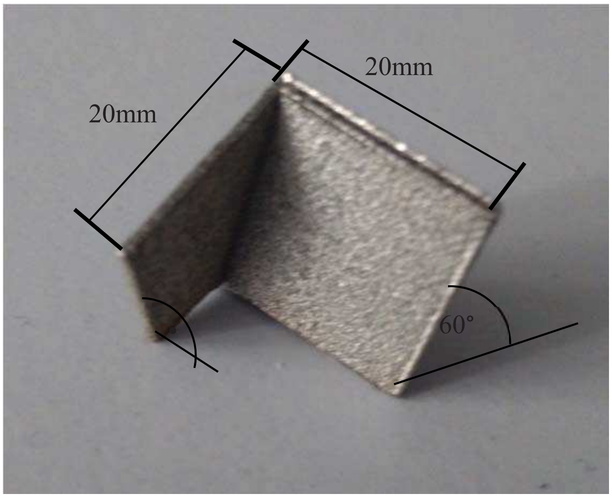


Fig. 2. Representative 316L stainless steel sample obtained during SLM process

Argon was used as the protective gas. Samples, after manufacturing, were cleaned in an ultrasonic scrubber to remove adhered powdery particles housed between print paths.

Structural studies and porosity measurements were performed on the Zeiss AxioVision material microscope. Dedicated software was used, based on the obtained images, the volume and the porosity of the pores in the material structure were calculated. Samples for structural investigation were electrolytically digested with 10% oxalic acid at 12 V and a current density of about 0.5 A/cm^2.

Vickers hardness measurement was performed on the Future Tech FM 700 Hardness Tester with ARS 9000 Automatic Acquisition System. A load of 5 N (0.5 kg) was applied and the load maintenance time was 15 s. A total of 10 measurements were made on each sample. Measurements were made according to the Vickers Hardness Test Standard PN-EN ISO 6507-1.

The measurements of the roughness of the inner and outer surfaces samples were carried out on the profilographometer Sutronic 25 of Tylor-Hubson Company. It was assumed the measurements length was 1.25 mm and measurement accuracy was ±0.01 μm. The parameter Ra acc. the Standard PN-EN ISO 4287:1999 was assumed as the quantity describing the roughness.

The qualitative and quantitative analysis of the chemical composition on the surface of the investigated samples was carried out using the X-ray energy dispersive spectroscopy (EDS) with the application of the spectrometer EDS LINK ISIS of the Oxford Company being a component of the electron microscope Zeiss Supra 35. The research was carried out with the accelerating voltage of 20 kV.

3. Results and discussion

Measurements of technological properties of powder SS 316L presenting that the flowability according to the PN-82/H-04935 standard was 22 s. Bulk density according to the PN-EN 23923-1:1998 standard was 4.19 g/cm^3 and tapped density according to the PN-EN ISO 3953:2011E standard was 4.92 g/cm^3.

Figure 3 shows the morphologies of the 316L stainless steel powders. As shown, the as-received SS 316L powder is dominantly regular with spherically shaped powders. Table 4 show chemical composition analysis of the area presented in Figure 3.

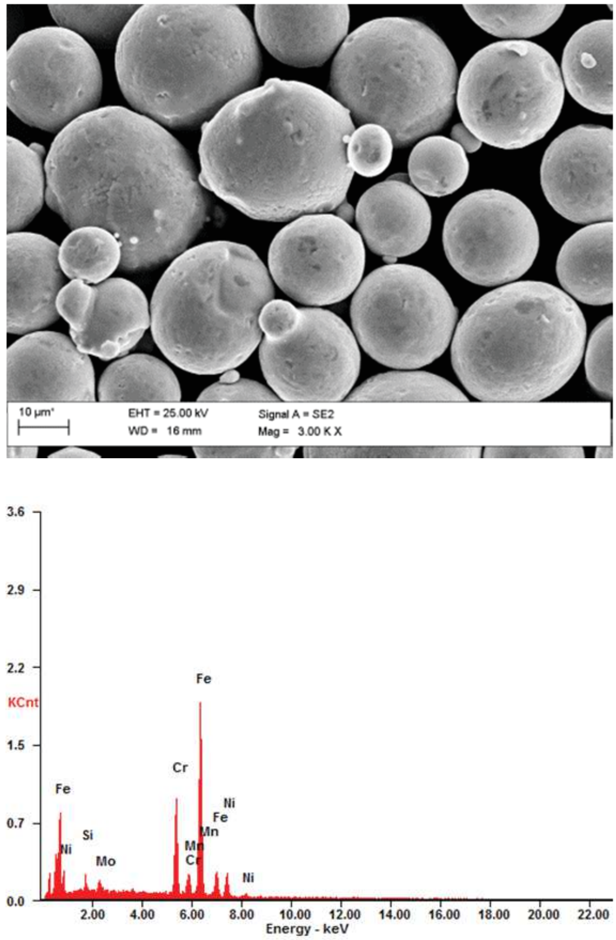


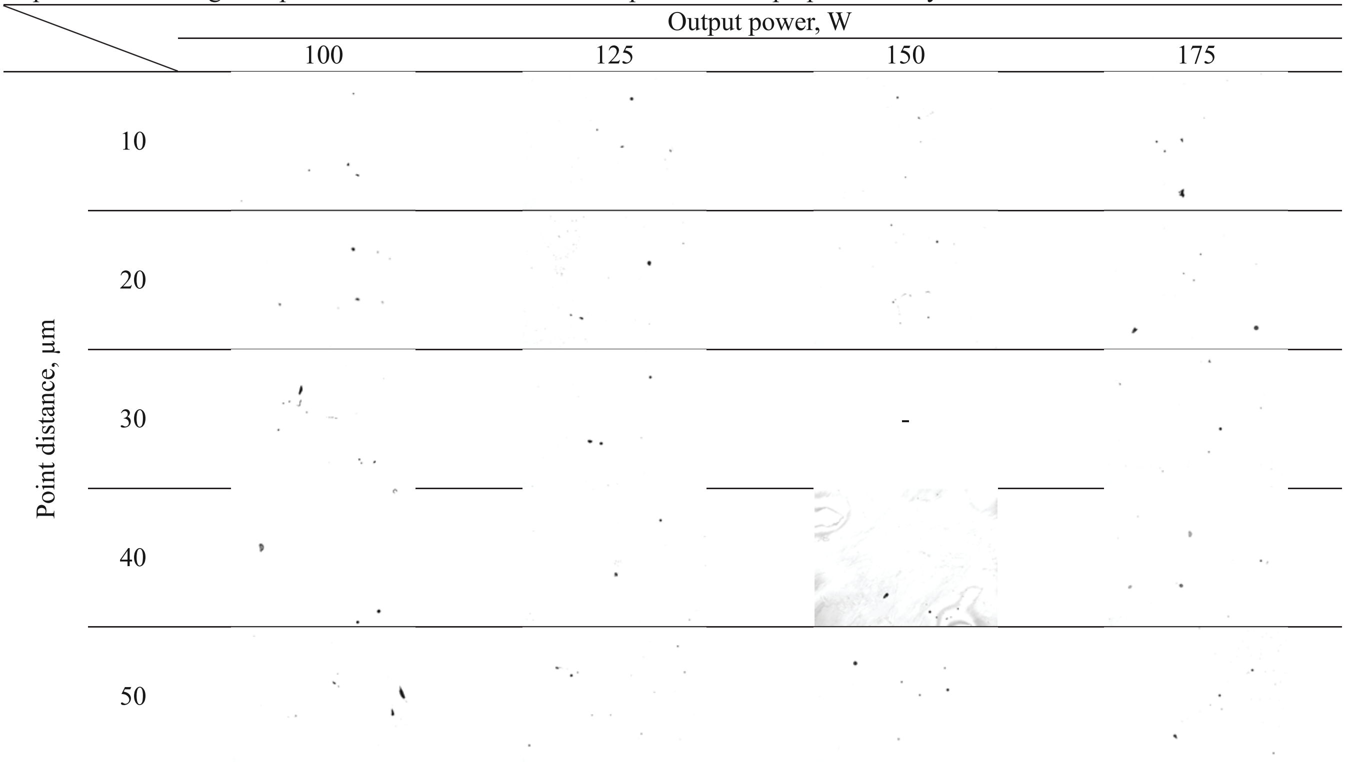
Fig. 3. Morphologies of the 316L stainless steel powders (SEM-SE) and its X-ray energy dispersive plot

Table 4.
Chemical composition analysis of the area presented in Figure 3

Element	wt.%	at.%
Si	1.6	3.1
Mo	2.6	1.5
Cr	19.8	21.0
Mn	3.1	3.1
Fe	61.7	60.8
Ni	11.0	10.3

The porosity measurements were taken to determine the average number of pores and the average surface area occupied by them. For all sample, five measurements were made in the regions characteristic for each of the walls of the analysed SLM samples, and representative sample images are shown in Tables 5 and 6. During the manufacturing process, one of the sample was damaged, i.e. sample made at laser power 150 W and point distance 30 μm. Therefore the results of this sample are not included.

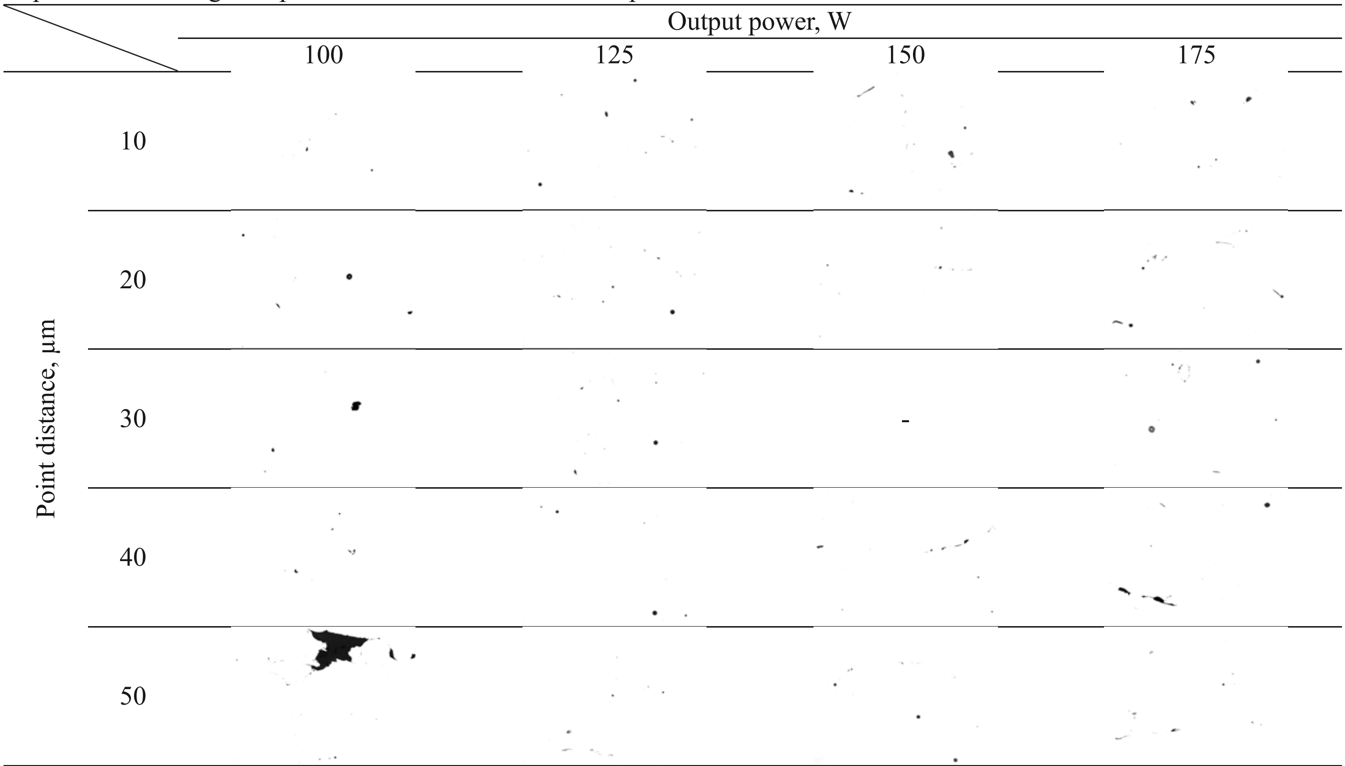
Table 5.
Representative images of pores in 316L stainless steel samples oriented perpendicularly to the substrate



Based on analysis of achieved structures can be observed that, depending on the laser power, point distance (i.e. energy density) and orientation of parts, the pore volume ratio varies. Part manufactured perpendicular to the substrate, low energy density and with 30 μm point distance, an average number of pores are significantly smaller. These are very different from the pores which occur in parts manufactured using energy density with high laser power and large point distance, where large caves and the discontinuities are formed. This is likely due to differences in porosity formation mechanisms.

Analysis of an average number of pores determined in parts perpendicularly oriented to the substrate shows that the lowest amount of pores was observed in part manufactured at 125 W laser power and point distance 30 μm and is slightly above 6 pores in the analysed areas, where energy density reached 174 J/mm^3. It can be assumed that for the manufacturing of parts from 316L steel at a 90° wall angle to the platform, these are optimum technological parameters to guarantee the lowest porosity of the obtained material and thin-walled products. The highest average pore count was noted for a sample manufactured at 150 W laser power and a 20 μm point distance (313 J/mm^3) where more than 35 pores in the tested area were observed. Analysis of average values of pores for parts manufactured with different laser power, can be stated that the smallest mean porosity is obtained for parts made at laser power 125 W, while the largest porosity achieved at 150 W. Similarly, by analysis of porosity as a function of the point distance, it can be noted that the lowest average porosity is obtained when manufacturing process is set at 30 µm, while the largest porosity for point distance equal to the 20 µm.

Table 6.
Representative images of pores in 316L stainless steel samples oriented at 60° relative to the substrate



Interpretation of results of an average number of pores in parts oriented at 60° to the platform, it can be stated that the lowest amount of pores was observed in part manufactured at 100 W laser power and point distance at 30 µm (139 J/mm^3) and is less than 4 pores on average in the analysed areas. It can be assumed that for the manufacturing the parts from 316L steel oriented at a 60° to the platform, these are optimum technological parameters to guarantee the lowest porosity of the obtained material and thin-walled products. The highest average pore count was recorded for a sample produced at 175 W laser power and a 40 µm (185 J/mm^3) point distance where 35 pores in the tested area were observed. By summarising and comparing the number of pores in analysed parts, it can be stated that the smallest amount of pores is characterised by manufactured samples at a point distance of 30 µm with a laser power of 125 W for parts oriented perpendicularly to the substrate and 100 W for elements oriented at 60° to the substrate. The average percentage area occupied by the pores (pore volume) is shown in Figure 4.

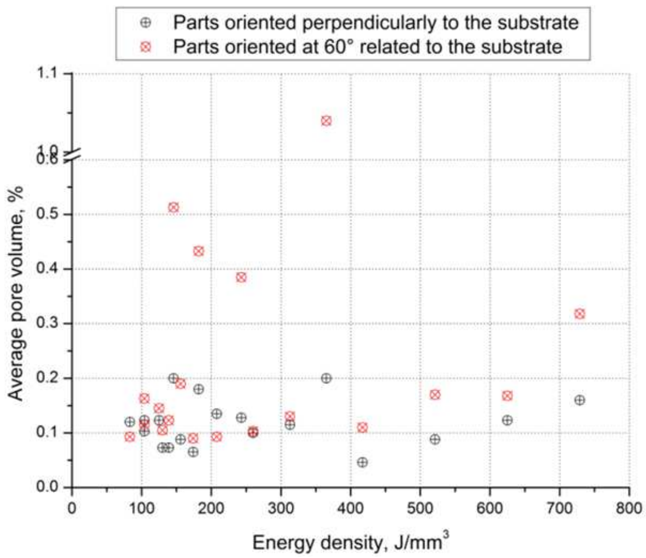


Fig. 4. Average pore volume in relation to the energy density of manufactured SLM parts

The figure display that in analysed micro-regions, the lowest average pore volume was observed in part manufactured at 400 J/mm^3 (100 W/10 μm), where is almost 0.05%. It can be assumed that for the manufacturing parts from 316L steel perpendicularly oriented to the substrate, these are optimum technological parameters to guarantee the lowest pore volume of the obtained material and thin-walled products. However, when energy density drops to approx. 150 J/mm^3 (175 W/40 μm), a noticeable 0.18% porosity could be observed. The lowest average pore volume of samples oriented at 60° relatives to the substrate was noted in a sample manufactured at 125 W laser power and with point distance 30 μm, and is amount about 0.09%. It can be assumed that for the manufacturing parts from 316L steel oriented at 60° relative to the substrate, these are optimum manufacturing parameters to guarantee the lowest porosity of the obtained material and thin-walled products. The highest average pore volume was achieved for a part manufactured at 175 W laser power and a with 10 μm point distance, and the amount was more than 1%. Analysis of average pore volume fabricated SLM parts as a function of laser power; it can be concluded that for all investigated parts, increases a laser power causes an increase in average pore volume. Moreover, changes in point distance for parts made at the same laser power does not cause changes in average pore volume.

The pores recognised in the examined samples are not spherical, ruling out that they are 'metallurgical' pores caused by entrapment of gas (hydrogen) in the melt. Since the pores are variable in shape, this implies the presence of keyhole pores that form when there is a keyhole instability. Various reasons may cause the formation of keyhole instabilities such as high cooling rate, insufficient energy density, improper laser offset at the power bed. Another purpose of increased porosity is reducing the overlapping area of two adjacent melt tracks by increases point distance.

Given that the optimum manufacturing parameters from the number of pores and their volume share for parts oriented perpendicularly and at 60° relative to the substrate differ from both analysed results, it can be stated that it is not possible to determine the recommended minimum printing parameters values criteria mentioned above. The task of an engineer preparing the technological parameters of printing elements and products with this technology will be the decision which of the requirements (quantity or share of pores) has in a given application more importance for the durability and quality of the material. It is also worth emphasising that no concentration of pore or number of pore concentrations was noticed, for example near the edges of the archived parts.

The results obtained from hardness test is presented in Figure 5. It can be concluded that for parts manufactured perpendicularly to the substrate, the lowest hardness was observed in a sample produced at 150 W laser power and with point distance at 10 μm, and reached 214 $HV_{0.5}$. The highest hardness was achieved for a part manufactured at 125 W laser power and a 10 μm point distance, where the hardness was 239 $HV_{0.5}$. It can be assumed that for the manufacturing of parts from 316L steel oriented perpendicularly to the substrate, these are optimum manufacturing parameters to guarantee the highest hardness of the obtained thin-walled products.

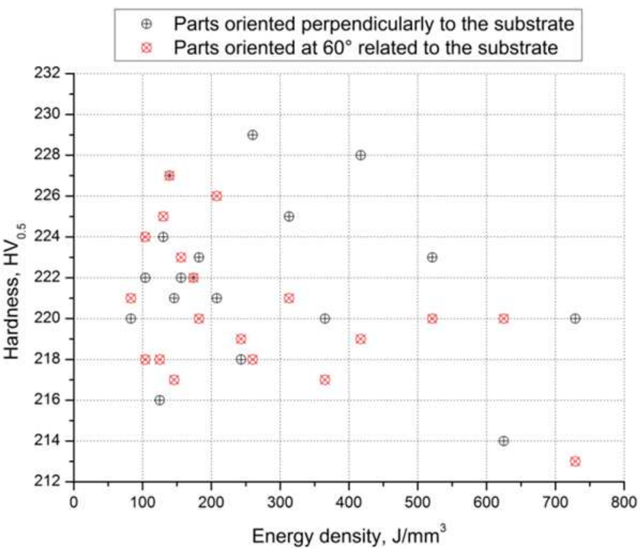


Fig. 5. Hardness [$HV_{0.5}$] of fabricated SLM at various energy density

Parts oriented at 60° to substrate manifested that the lowest hardness was achieved in SLM part produced at 175 W laser power and point distance 10 μm, and it was 213 $HV_{0.5}$. The highest hardness was achieved for SLM part produced at 100 W laser power and a 30 μm point distance, and it was 227 $HV_{0.5}$. Analysis of hardness of fabricated SLM parts as a function of laser power it can be noted that, increases in laser power cause decrease in hardness of investigated SLM parts. This tendency is caused by increases in the average pore volume of analysed SLM parts, i.e. increasing laser power cause increase in average pore volume thereby decrease in hardness.

The light optical microscope was used for microstructural observation of as-built parts after etching. Tables 7 and 8 show an array of micrographs of parts fabricated using various laser power and point distance.

Table 7.
Representative microstructure of parts oriented perpendicularly to the substrate

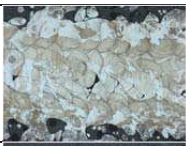
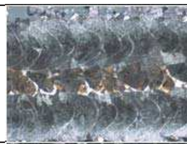
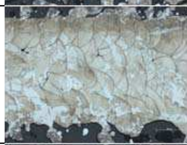
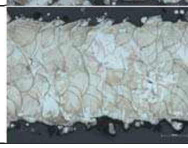
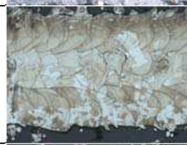
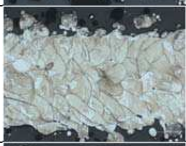
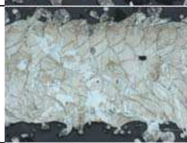
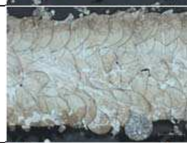
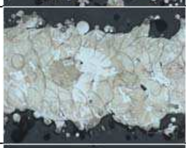
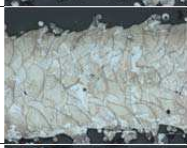
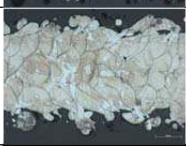
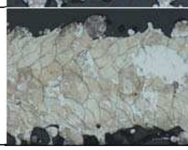
Point distance, µm	Output power, W: 100	125	150	175
10				
20				
30			-	
40				
50				

Table 8.
Representative microstructure of parts oriented at 60° relative to the substrate

Point distance, µm	Output power, W: 100	125	150	175
10				
20				
30			-	
40				
50				

Based on analysis of structures of the SLM parts with different power laser and the variable point distance, was found that increased as the laser power, the degree of melting of the powder grain was raised in the structure of the material after a process. Comparing the effect of point distance on structure, it was observed that for 20 μm values, the structure was characterised by the most significant orderliness and regularity of the degree of remelting. For this point distance, the structure has a regular stripe pattern. For other point distance values, this balance was more or less disturbed. When analysing the effect of the print angle of the examined surface relative to the base plate, it was found that for 60° angles, the structure is severely disordered and strongly differentiated regarding the grain elongation direction within the sample cross-section. Such phenomena were not observed for parts oriented perpendicularly. Most often, for an angle of 60°, the outer grain is elongated in the direction of printing (incremental production) and the centre part of the sample axis across the print direction. The cause of such phenomena is probably another angle of incidence of the beam on the powder layer, which causes such variation. Nevertheless, it is possible to point to a 90° angle of view that is much more favourable for microstructure and homogeneity, which will also provide less anisotropy the material obtained. The grain size of the structures obtained is quite varied within one sample and correlated with the grain size of the powder used in this case for printing. Observing the structures obtained at 90 and 60°, it is difficult not to resist the impression confirmed in the above tests that the porosity of 60° printed samples is greater and contains more extensive pores than oriented perpendicularly.

Tables 7 and 8 show the typical macrostructure of 316L steel fabricated using SLM at lower magnification. The overlapping, bowl-shaped features on cross-sections are formed in the melt pool solidification process. These melt pool features, which are created by each laser scan, are parallel to the build direction. The columnar grains exhibit well-favoured orientation following they develop simultaneously with the heat transfer directions.

The scanning tracks exhibit the unusual wave feature of SLM process. There exists trench within two adjacent contiguous tracks, and also exists ridge for single track, which confirms that the inter-layer staggered scanning strategy is beneficial to improve the SLM fabricated part's density. The inter-layer staggered joint effect between two adjacent layers can be clearly observed. The melting and joint effect between two layers are excellent.

The different surface quality can be observed by the roughness measure, as shown in Figure 6. The surface roughness of the SLM parts orientated at 60° to the substrate is apparently lower than that surface of the models orientated perpendicularly to the substrate. Thus, it can be considered that the deep holes and adherent semi-melted particles on the surface strongly increase the surface roughness and decrease the dimensional accuracy.

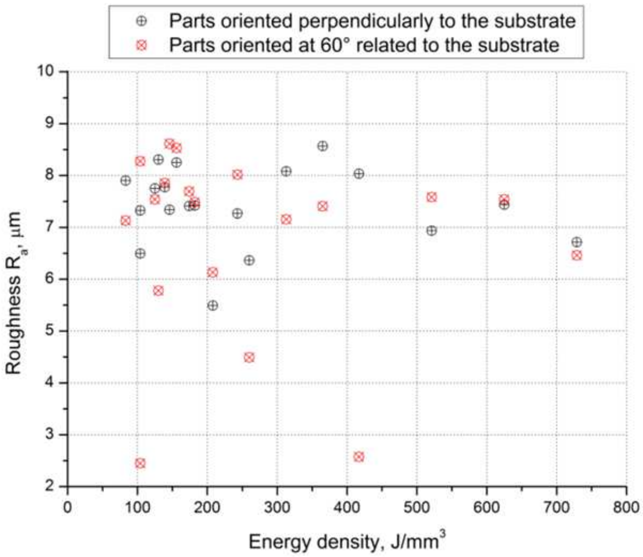


Fig. 6. Influence of energy density on roughness

The lowest value of roughness the surfaces oriented at an angle of 60° relatives to the substrate was noted for model performed by laser power output 125 W and the point distance 50 μm was 2.45 μm. The highest value pointed out for the roughness of the model made by laser power output 175 W and point distance 20 μm where the value of this parameter was 9.1 μm.

The lowest roughness of part oriented perpendicular to the substrate was observed for the wall made by laser power output of 100 W and the distance point 20 μm was 5.49 μm, and the highest value noted for the roughness of the wall formed by laser power output of 150 W and a point distance of 20 μm where the value of this parameter was 8.59 μm.

4. Conclusions

The applicability of the applied energy density equation through the SLM process has been studied for 316L parts. The 316L SLM parts were developed utilising various process parameters i.e. laser power and point distance. The 20 test parts oriented perpendicularly and at 60° relatives to the substrate, using 316L stainless steel powders have been successfully manufactured using an AM125 Renishaw machine. Numerous experiments and analysis have been done to the as-built test parts to determine the influence of energy density, a critical factor in the selective laser melting process, on porosities, microstructures, roughness and hardness of as-built parts.

The results can be summarised as follows:

- Porosity test showed that to a part oriented perpendicular to the platform, the highest number of pores with a relatively low percentage of surface area occupied by them were obtained for samples produced with a laser power of 150 W, and the point distance 20 μm (313 J/mm^3). For this wall, the lowest amount of pores was obtained for a sample produced at a 125 W laser power and a point distance of 30 μm (174 J/mm^3). Parts oriented at 60° to the platform are characterised the lowest pore volume, i.e. 100 W sample and a 30 μm point distance (139 J/mm^3).
- Hardness studies have shown that, for the majority of sample parameters taken, they had a higher hardness on the parts oriented perpendicular to the substrate than oriented at 60° relative to the substrate. This may be due to better adhesion of powdered particles perpendicular to the substrate. The highest hardness up to 233 $HV_{0.5}$ was obtained for a part perpendicular to the substrate produced at a 125 W laser power and a point distance of 20 μm.
- Structural observations have shown that with the changes in energy density, the grain size of the analysed material after SLM process is changing. The higher laser power and the lower the point distance cause a decrease in grain size as a result of rapid melting and solidification cycles.

The surface quality parts using 316L steel with the assumed parameters of the experiment depends on the process parameters used during the SLM technique as well as the orientation of formed walls of the model relative to the substrate and thus the laser beam. The lowest roughness of 316L SLM parts oriented perpendicularly to the substrate was found when 100 W and 20 μm the distance point was utilised. The lowest roughness for part oriented at 60° relatives to the substrate was observed when 125 W and the point distance 50 μm was employed.

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